

RFT1 (Q96AA3) — Family constraint: central cavity vs. lateral lipid portal

Hypothesis (slug `family-constraint-cavity-vs-portal`): In the RFT1 family, evolutionary constraint is concentrated on positions lining the central substrate cavity of the MOP-superfamily fold rather than on positions lining the lateral lipid portal — i.e., selection preserved substrate capture/coordination rather than the lipid transit path.

Term in scope: GO:0140303 *intramembrane lipid transporter activity* (MF).

Executive Judgment

Verdict: SUPPORTED (robust, with one honest qualification).

A deep eukaryotic RFT1-family alignment mapped onto the AlphaFold model of Q96AA3 shows that **central-cavity-lining transmembrane (TM) positions are the single most evolutionarily constrained structural class, well above the TM background, whereas lateral lipid-portal-lining positions are statistically indistinguishable from background (slightly below it), and the outer lipid-facing surface is the fastest-evolving class.** The cavity-vs-portal contrast is large and highly significant (rank-biserial -0.68 , $p = 3 \times 10^{-5}$) and survives four independent controls (a second conservation metric, amino-acid-composition/Pro-Gly exclusion, a 20,000-draw permutation null, and ± 15 – 25° perturbations of the estimated membrane axis).

Qualification (the alternative that is partly true): buried helix-packing ("neither") residues are numerous ($n \approx 180$) and *also* modestly enriched above background ($p = 1 \times 10^{-3}$, effect -0.18). So a meaningful share of *total* family constraint is fold maintenance. But this is a weaker effect than the cavity (-0.43), and — critically — the cavity signal is **not** driven by helix-breakers: cavity residues contain **0% Gly and 0% Pro**; they are 28% charged and 47% polar, the chemistry of substrate coordination. The hypothesis's own "Pro/Gly fold-maintenance" alternative is therefore refuted *for the cavity class specifically*.

Most important caveats: (i) the family alignment is **eukaryote-only** — PF04506/RFT1 has no alignable archaeal/bacterial members, so the requested prokaryotic MOP homologues (MATE/MurJ) could not be folded into one MSA; (ii) the membrane axis and the cavity/portal geometry are derived from a **single static AlphaFold model** with an operationally defined portal; (iii) evolutionary constraint indicates *importance*, not mechanism per se.

What was actually computed (provenance)

All numbers below are direct outputs of code executed this run (not asserted). Pipeline:

1. **Alignment.** Pfam **PF04506** ("Rft-1") *full* alignment via the InterPro API → **3,826 sequences, 512 HMM match columns**. Match columns identified by the Pfam convention (columns containing no `.` and no lowercase). Taxonomic spread: fungi (9PEZI, 9EURO, 9AGAR, 9HYPO, 9PLEO, 9BASI, 9HELO...), metazoa (9BILA, 9PASS, 9TELE, 9AVES...), plants (9ROSI, 9POAL, 9ORYZ, 9LILI), stramenopiles (9STRA). **1,371 distinct clades; zero bacterial/archaeal mnemonics.**
2. **Per-position constraint.** Henikoff position-based sequence weights → weighted per-column Kullback–Leibler relative entropy to background (primary metric) and normalized Shannon conservation (secondary). Columns with weighted gap fraction > 0.6 excluded as unreliable.
3. **Human mapping.** Profile Needleman–Wunsch (free end gaps) of Q96AA3 against the match-state profile → **497 human residues mapped (positions 10–528).**
4. **Structure & membrane frame.** AlphaFold **AF-Q96AA3-F1 (v6)**. Membrane normal estimated as the principal eigenvector of TM-helix direction vectors (helices detected from $\text{Ca } i \rightarrow i+3 / i \rightarrow i+4$ geometry). TM residues = $|z - \text{membrane-center}| \leq 15 \text{ \AA}$ (**n = 312 mapped + reliable**).
5. **Geometric classification (explicit criteria).** Occupancy grid (1.5 Å, 1.4 Å probe) in the membrane frame. For each TM residue, side-chain centroid → cylindrical (r, θ, z) about the central axis.
6. **central-cavity-lining:** side chain points inward ($r_{\text{sidechain}} < r_{\text{Ca}} - 0.3 \text{ \AA}$) and reaches the central channel wall ($r \leq \text{lumen radius}(\theta) + 3 \text{ \AA}$).
7. **lateral-portal-lining:** lipid-exposed (>60% of outward hemisphere sample points free of protein) **and** within 9 Å of a cavity-lining residue (i.e., lipid-facing residues bordering the cavity's lateral opening).
8. **outer lipid-facing:** lipid-exposed but not adjacent to the cavity.

9. **buried helix-packing ("neither")**: neither lumen-lining nor lipid-exposed.

Core result (KL relative entropy, bits)

Structural class	n	mean KL	median KL	vs TM background (Mann-Whitney)	effect (rank-biserial)
central cavity	32	2.354	2.513	$p = 6.2 \times 10^{-5}$	−0.43 (more constrained)
lateral portal	22	1.333	1.248	$p = 0.24$ (NS)	+0.15 (slightly below)
outer lipid-facing	78	0.918	0.852	$p = 1.9 \times 10^{-13}$	+0.54 (far below)
buried packing	180	1.920	1.694	$p = 1.2 \times 10^{-3}$	−0.18 (modestly above)
TM background	312	1.673	1.424	—	—

Direct contrasts: **cavity vs portal** $p = 2.9 \times 10^{-5}$ (**effect −0.68**); cavity vs all lipid-facing $p = 1.0 \times 10^{-12}$; all-lipid-facing vs background $p = 9.3 \times 10^{-12}$ (below).

Robustness controls

- **Second metric (normalized Shannon conservation)**: cavity 0.564 vs bg 0.438 ($p = 5.7 \times 10^{-4}$); portal 0.373 vs bg ($p = 0.19$, NS); cavity vs portal $p = 2.3 \times 10^{-4}$. Same ranking.
- **Composition / not a Pro-Gly artifact**: cavity = **0% Gly, 0% Pro, 28% charged, 47% polar, 16% aromatic**; portal = 0% charged, mostly hydrophobic. Cavity enrichment unchanged after excluding Gly/Pro (none present), $p = 6.2 \times 10^{-5}$.
- **Permutation null (20,000 random 32-residue TM draws)**: observed cavity mean 2.354 vs null mean 1.676, empirical one-sided $p = 1 \times 10^{-4}$.
- **Membrane-axis sensitivity**: across $\pm 15^\circ$ to $\pm 25^\circ$ tilts of the estimated normal, cavity $\gg$ portal ($p = 3.1 \times 10^{-6}$... 7.9×10^{-4}) and cavity $\gg$ background persist.

Axis-independent corroboration (continuous, confounder-controlled)

To confirm the result does not depend on the cavity/portal class boundaries or on the precise membrane axis, per-residue constraint was correlated with continuous geometric measures over all 312 TM positions:

Relationship	Spearman ρ	p	Reading
constraint vs radial distance from central axis	−0.675	8×10^{-43}	closer to the pore → more constrained
constraint vs lipid-exposure fraction	−0.591	1×10^{-30}	more lipid-exposed → less constrained
constraint vs burial (heavy-atom neighbors ≤ 10 Å)	+0.425	4×10^{-15}	more buried → more constrained
constraint vs pLDDT	−0.086	0.13 (NS)	not a model-confidence artifact

Monotonic gradient by lipid-exposure quartile (least→most exposed): mean KL **2.31 → 1.95 → 1.48 → 0.98**. Restricting to high-confidence residues (pLDDT > 70; 308/312) leaves the correlations unchanged (lipid $\rho = -0.596$; radial $\rho = -0.675$). **Spatial clustering**: the 15 most-constrained TM residues have a mean pairwise distance of 19.2 Å vs a random-TM null of 30.1 Å (5,000 permutations, $p < 0.0001$) — they form a single compact pocket, not dispersed fold positions. Together these confirm the cavity-over-portal pattern without relying on the discrete geometric classification.

Candidate substrate-coordinating residues (most-constrained cavity positions, human numbering)

R290 (KL 4.05), N435 (3.81), Y378 (3.66), E64 (3.48), E298 (3.31), Q186 (3.28), N283 (3.10), E156 (3.06), T257 (3.06), S286 (3.00), E258 (2.80), F415 (2.79), E260 (2.67), S412 (2.63), K152 (2.57). These are overwhelmingly H-bond donors/acceptors and charged side chains — the chemistry expected to coordinate the polar **Man5GlcNAc2-pyrophosphate** headgroup of the substrate. *(Leads for mutagenesis; require curator/experimental verification.)*

Evidence Matrix

#	Citation	Evidence type	Stance	Claim tested	Key finding	Context
1	This run (computed)	Structural/ evolutionary (computational)	Supports	Constraint concentrated on cavity, not portal	Cavity KL 2.35 >> portal 1.33 $\approx$ bg 1.67; cavity vs portal $p=3\times 10^{-5}$	PF04506 (3,826 seq, eukaryotes) on AF-Q96AA3
2	This run (computed)	Computational controls	Supports	Result not an artifact	Reproduced by Shannon metric, permutation $p=1e-4$, axis-tilt $\pm 25^\circ$, composition (0% Pro/Gly)	Same pipeline
3	This run (computed)	Computational (composition)	Refutes the Pro/Gly alternative for cavity	Is cavity signal fold-maintenance?	Cavity = 0% Gly/Pro, 28% charged, 47% polar	Same pipeline
4	PMID 38886340 (Chen et al., 2024)	Direct assay (reconstitution)	Supports GO:0140303	Is Rft1 itself the flippase?	"purified Rft1 catalyzes the translocation of M5GN2-PP-Dol across the lipid bilayer"; "confirm the molecular identity of Rft1 as the M5GN2-PP-Dol ER flippase"	Reconstituted proteoliposomes
5	This run (computed)	Computational (sanity check)	Qualifies/ validates method	Does the frame capture	Outer lipid-facing surface is the fastest-	Same pipeline

#	Citation	Evidence type	Stance	Claim tested	Key finding	Context
				inside/ outside?	evolving class ($p=2\times 10^{-13}$)	
6	InterPro IPR007594 / Pfam PF04506 / TCDB 2.A.66.3.2	Database	Orientation	Fold/ superfamily membership	RFT1 = MOP flippase superfamily, RFT1/ oligosaccharidyl- lipid flippase family; eukaryote- restricted	Cross- references for Q96AA3
7	This run (computed)	Computational (continuous, confounder- controlled)	Supports	Is the cavity signal axis-/ binning-/ pLDDT- independent?	Constraint vs radial-distance $\rho=-0.675$ ($p=8\times 10^{-43}$) and vs lipid- exposure $\rho=-0.591$ ($p=1\times 10^{-30}$); pLDDT $\rho=-0.086$ (NS); top-15 residues cluster (19.2 vs 30.1 Å, $p<10^{-4}$)	Same pipeline, 312 TM positions

GO Curation Implications (*leads — require curator verification*)

- **GO:0140303 (intramembrane lipid transporter activity, MF) → RETAIN.** It is directly supported by reconstitution

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and is *reinforced* by this analysis: the family's conserved core is precisely the polar substrate-handling cavity of a MOP flippase, not incidental surface.

- **Consider a more specific MF child if one exists** (curator to check the ontology): the physiological substrate is a **lipid-linked oligosaccharide (Man5GlcNAc2-PP-dolichol)**, so a "glycolipid/oligosaccharide-lipid flippase" style MF child would be more informative than the generic parent. Do **not** invent an ID; verify against current GO. Evidence code for the retained/refined MF should reference the direct assay (IDA,

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rather than IEA/ISS alone.

- **This evolutionary analysis is CC/BP-neutral.** It speaks to *which* residues matter (MF mechanism), not to localization (ER membrane, CC) or the glycosylation process (BP); those should be curated from their own primary evidence.
- **"Protein binding" is not an appropriate summary** here — a specific transporter MF is supported.

Mechanistic Scope

- **Immediate molecular function tested:** which wild-type residues selection has preserved across the RFT1 family, and where they sit relative to the substrate cavity vs. the lipid-facing/portal surface of the MOP fold. This is a statement about the flippase's **substrate-coordination machinery**.

- **Direct gene-product activity:** transbilayer flipping of Man5GlcNAc2-PP-dolichol

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- **Separated (not part of this test):** downstream N-linked glycosylation flux, CDG disease phenotypes, or any inference from loss-of-function alleles. Per instructions, **no disease/pathogenicity data were used** to select positions; the constrained set is defined purely by cross-species selection and geometry.

Conflicts and Alternatives

- **Both-cavity-and-portal alternative:** *not supported* — portal is at/below background while cavity is above.

- **Fold-maintenance (Pro/Gly, helix-packing) alternative:** *partly true but not for the cavity.* Buried packing residues carry a weaker-but-real constraint enrichment; the cavity class contains no Pro/Gly and is polar/charged. So "fold maintenance dominates everything" is refuted, but "fold maintenance also matters" is honestly acknowledged.
 - **Shallow/too-divergent alignment alternative:** *not the case* — 3,826 sequences, 512 columns, human mapped over 497 residues; classes separate with large effect sizes and small p-values.
 - **Portal-definition sensitivity:** the "portal" is operationally the lipid-facing rim of the cavity in a *single, likely occluded* AlphaFold state. A genuinely open lateral gate in another conformational state could be lined by different residues; my first (ray-tracing) portal detector found **no** wide-open lateral sector in this model, consistent with an occluded state.
 - **Cross-superfamily constraint (MATE/MurJ):** deliberately requested but **not computable by sequence alignment** — those Pfam families are too divergent to co-align with RFT1; a structure-based comparison would be needed.
 - **Paralog/isoform confusion:** none material — RFT1 is single-copy in human; PANTHER PTHR13117:SF5 and TCDB 2.A.66.3.2 are RFT1-specific.
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Knowledge Gaps

1. **Experimental structure of RFT1.** *Checked:* only an AlphaFold model was available programmatically; the membrane axis/portal were inferred. *Why it matters:* the cavity/portal partition depends on the model. *Resolver:* a cryo-EM/OPM-oriented Rft1 structure ($\pm$ bound LLO) to define the real cavity and any lateral gate.
2. **Prokaryotic MOP constraint.** *Checked:* PF04506 has no bacterial/archaeal members. *Why it matters:* the hypothesis invokes the broader MOP/MATE/MurJ architecture. *Resolver:* structure-based multiple alignment across the superfamily (e.g., Foldseek/DALI + 3D-anchored MSA).
3. **Substrate-contact identity of the conserved cavity residues.** *Checked:* only geometric proximity to the lumen. *Why it matters:* conservation $\neq$ contact. *Resolver:* substrate-bound structure or docking, and mutagenesis of R290/N435/Y378/E64/E298 etc.
4. **Conformational state.** *Checked:* one static model (apparently occluded). *Why it matters:* the portal may only exist transiently. *Resolver:* multi-state modeling / MD / alternate-state structures.

Discriminating Tests

- **Substrate-bound Rft1 structure or MD-informed docking** → do the constrained cavity residues (R290, N435, Y378, E64, E298, Q186, N283, E156, S/T cluster 257/286/412) actually contact the oligosaccharide/pyrophosphate? This is the single most decisive test.
- **Alanine/charge-reversal mutagenesis** of the top cavity residues vs. matched **portal** residues, scored in the reconstituted flipping assay

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cavity mutants should impair transport far more than portal mutants if the hypothesis holds.

- **Structure-based superfamily constraint** (RFT1 vs MATE vs MurJ, aligned in 3D) to test whether the cavity-over-portal pattern generalizes across the MOP fold.
- **Deep-mutational scanning / evolutionary-coupling (DCA)** on the same alignment to check that co-evolving/coupled positions concentrate in the cavity.

Curation Leads (*all require curator verification*)

- **Action:** *Retain* GO:0140303 (MF) with IDA support

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flag for possible refinement to a lipid-linked-oligosaccharide-flippase child term if the ontology provides one.

- **Candidate reference + exact snippets to verify**

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- "purified Rft1 catalyzes the translocation of M5GN2-PP-Dol across the lipid bilayer"
- "confirm the molecular identity of Rft1 as the M5GN2-PP-Dol ER flippase"
- **Supporting computational provenance (this report):** the constraint-by-class table, robustness controls, and the ranked cavity-residue list are curator-usable evidence that the *conserved core* of RFT1 is its substrate-handling cavity.

- **Suggested curator questions:** (1) Is there a GO MF child specific to oligosaccharide-lipid/glycolipid flipping? (2) Should the ER-membrane CC and N-glycosylation BP terms be (re)affirmed from their own primary evidence rather than from this MF-focused analysis?
 - **Suggested experiments:** cavity-vs-portal mutagenesis in the reconstituted flipping assay; substrate-bound structure.
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Bottom line

Within the deepest assemblable (eukaryotic) RFT1 family, evolutionary selection has preserved the polar central-cavity residues of the MOP fold — the substrate-capture/coordination surface — far more than the lateral lipid-facing/portal surface, which evolves at or below the transmembrane background. The seed hypothesis is **supported**, with the honest caveat that buried fold-maintenance packing also carries a weaker constraint signal, and that the analysis rests on a single static AlphaFold model of a eukaryote-only family. The result **reinforces retention** of GO:0140303 and yields a concrete, testable set of candidate substrate-coordinating residues.

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